

An Introduction *to* Hyperledger Fabric

With the runaway success of Bitcoin, the interest in distributed ledgers and blockchains has perked up. Hyperledger Fabric is a collaborative open source project that has been created to advance cross-industry blockchain technologies. This article introduces the reader to it.

Hyperledger Fabric was the first proposal to The Linux Foundation's Hyperledger Project, contributed in part by IBM. This framework is modular, scalable and secure. It has been specifically designed for industries such as manufacturing, agriculture, healthcare and the capital markets. Hyperledger Fabric is revolutionary in its ability to allow entities to conduct confidential transactions without passing through a central authority. It is unique, because it allows for modular consensus and membership service.

There are four different types of roles within a Hyperledger Fabric network:

- **Clients** are applications that act on behalf of a person to propose transactions on the network.
- **Peers** maintain the state of the network and a copy of the ledger. There are two different types of peers — endorsing and committing peers. However, there is an overlap between endorsing and committing peers, in that the former are a special kind of committing peers. All peers commit blocks to the distributed ledger.
- **Endorsers** simulate and endorse transactions.
- **Committers** verify endorsements and validate transaction results, prior to committing transactions to the blockchain.

The ordering service accepts endorsed transactions, orders them into a block, and delivers the blocks to the committing peers.

Key components of Hyperledger Fabric

Transaction endorsement

A transaction endorsement is a signed response to the results of the simulated transaction. The method of transaction endorsements depends on the endorsement policy, which is specified when the chaincode is deployed. An example of an endorsement policy would be something like, "The majority of the endorsing peers must endorse the transaction." Since an endorsement policy is specified for a specific chaincode, different channels can have different endorsement policies.

Ordering

Transactions within a time frame are sorted into a block and are committed in sequential order. In a blockchain network, transactions have to be written to the shared ledger in a consistent order. The order of transactions has to be established to ensure that the updates to the world state are valid when they are committed to the network. Unlike the Bitcoin blockchain, where ordering occurs by the solving of a cryptographic puzzle or by mining, Hyperledger Fabric allows the organisations running the network to choose the ordering mechanism that best suits that network. This modularity and flexibility makes Hyperledger Fabric incredibly advantageous for enterprise applications.